

Phase 7: Operations & Process - Pit Stop Time Variation (DMAIC)

For this phase, we will be applying a Lean Six Sigma lens to the F1 pit stop timing data.

DMAIC

- Define: Variations in pit stop leads to losing track position. Teams analyse a lot of telemetry data and gaps between drivers to decide when to pit their drivers. They plan these pit stops based on the pit stop time estimates. So if any unexpected variation occurs during the pit stop, all the data analyses render useless and the drivers lose track position.
- Measure: We pulled the pit stop data from FastF1. The first list we pulled sorted drivers alphabetically and then within drivers it sorted the races alphabetically. The list showed pit stop time variation for drivers within the same races. Variation among different races is understandable due to different pit entry lengths, but a huge variation within the same race raised suspicion. For example, we can see a jump from 119 seconds to 128 seconds for Alex Albon in Bahrain.
- Analyze: We pulled the mean, std and count data for all drivers in the next list. We sorted it by largest standard deviation values. Esteban Ocon had the highest variation of 22.38 among the 6 pit stops he had made. He was followed by Logan Sargeant with a std variation of 18.31 and Lewis Hamilton with 17.03. The most likely explanation for major variations within the same races has to be crew/driver/machinery error. Crew error could be bringing the wrong set of tires; Driver error could be incorrect parking; Machinery error could be the car not switching from neutral or the guns being stuck. The possibilities of errors are endless. I noticed a very unusual similarity between Alex Albon and Pierre Gasly's Bahrain pit stops. Both of them had 3 pit stops where the initial 2 were around the 120 sec mark but on their third pit stop, both jumped to around 128 seconds. This made me wonder if as the laps go on, due to Bahrain's heat the machinery faces issues? But looking at the other drivers' pit timings pattern did not align with this. While checking Monaco's pit data, it was evident that this track had the most variation for pit stops among all drivers. Monza had the least number of pit stops and time variation. It's hard to form any conclusions about what the definite reasons might be.
- Improve: A few evident and obvious ways to improve the process throughout a season would be to standardize the process understanding among all the crew members, regular machinery checks and flagging any outlier pit stops for detailed reviews.
- Control: Maintain a rolling average of pit stop times across the season to verify the standardizations and machinery checks are improving the phase. Another way is to establish a post race feedback protocol where any pit stop which exceeded the target time threshold is reviewed against the telemetry data to prevent any recurring crew or driver errors.